MENG WANG

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SUMMARY

I am a Ph.D. candidate in Computer Engineering at the University of British Columbia, studying how to design **scalable and efficient quantum computing systems**. My research focuses on **quantum error correction and the architectural and software techniques required for fault-tolerant quantum computing**, together with system-level optimizations across the quantum computing stack.

My work has appeared in top-tier computer architecture venues including **ISCA**, **MICRO**, and **ASPLOS**, where I have developed techniques for quantum circuit simulation, distributed and multi-device execution of quantum workloads, and circuit and compiler optimizations that reduce the cost of executing quantum programs, including those targeting **fault-tolerant, error-corrected systems**. These contributions advance quantum–classical co-design and help bridge the gap between theoretical quantum algorithms and practical large-scale quantum computing.

I also conducted research at Pacific Northwest National Laboratory (PNNL), applying high-performance computing techniques to quantum simulation challenges in collaboration with researchers in chemistry and physics.

EDUCATION

Ph.D in Electrical and Computer Engineering — *September 2020 - Present*
The University of British Columbia — Vancouver, BC, Canada
Advisor: Prof. Prashant J. Nair

BASc in Computer Engineering — *June 2020*
The University of British Columbia — Vancouver, BC, Canada
Graduated with Distinction

CONFERENCE PUBLICATIONS

1. **ISCA '26** — **Meng Wang**, Chenxu Liu, Samuel Stein, Yufei Ding, Poulami Das, Prashant J. Nair, and Ang Li
   Transpiler-Architecture Co-Design to Curb Clifford Costs in Fault-Tolerant Quantum Computing
   The International Symposium on Computer Architecture, 2026
2. **ISCA '25** — **Meng Wang**, Swamit Tannu, and Prashant J. Nair
   Accelerating Simulation of Quantum Circuits under Noise via Computational Reuse
   The International Symposium on Computer Architecture, 2025
3. **MICRO '24** — **Meng Wang**, Poulami Das, and Prashant J. Nair
   Qoncord: A Multi-Device Job Scheduling Framework for Variational Quantum Algorithms
   IEEE/ACM International Symposium on Microarchitecture, 2024
4. **ASPLOS '24** — **Meng Wang**, Bo Fang, Ang Li, and Prashant J. Nair
   Red-QAOA: Efficient Variational Optimization through Circuit Reduction
   International Conference on Architectural Support for Programming Languages and Operating Systems, 2024

JOURNAL PUBLICATIONS

1. **PRR '26** Sean R. Garner, Nathan M. Myers, **Meng Wang**, Samuel Stein, Chenxu Liu, and Ang Li
   Simulation of Thermal-Relaxation Noise for Quantum Error Correction
   Physical Review Research, **8(2)**, 023134, 2026

ARXIV PREPRINTS

1. **Meng Wang**, Chenxu Liu, Sean Garner, Samuel Stein, Yufei Ding, Prashant J. Nair, and Ang Li
   Tableau-Based Framework for Efficient Logical Quantum Compilation
   arXiv preprint arXiv:2509.02721
2. Aaron Hoyt, **Meng Wang**, Fei Hua, Chunshu Wu, Chenxu Liu, Muqing Zheng, Samuel Stein, Drew Rebar, Yufei Ding, Travis S. Humble, and others
   QASMTrans: An End-to-End QASM Compilation Framework with Pulse Generation for Near-Term Quantum Devices
   arXiv preprint arXiv:2602.05154
3. Chenxu Liu, **Meng Wang**, Samuel A. Stein, Yufei Ding, and Ang Li
   Quantum Memory: A Missing Piece in Quantum Computing Units
   arXiv preprint arXiv:2309.14432
4. Avinash Kumar, **Meng Wang**, Chenxu Liu, Ang Li, Prashant J. Nair, and Poulami Das
   Context Switching for Secure Multi-programming of Near-Term Quantum Computers
   arXiv preprint arXiv:2504.07048
5. Sean Garner, Chenxu Liu, **Meng Wang**, Samuel Stein, Ang Li
   STABSim: A Parallelized Clifford Simulator with Features Beyond Direct Simulation
   arXiv preprint arXiv:2507.03092
6. Zeguan Wu, Mingze Li, Muqing Zheng, **Meng Wang**, Junyu Liu, Samuel Stein, Ang Li, Yousu Chen, and Chenxu Liu
   Benchmarking and Resource Analysis for Augmented-Lagrangian Quantum Hamiltonian Descent
   arXiv preprint arXiv:2605.12066
7. Adrian Harkness, Shuwen Kan, Chenxu Liu, **Meng Wang**, John M. Martyn, Shifan Xu, Diana Chamaki, Ethan Decker, Ying Mao, Luis F. Zuluaga, and others
   FTCircuitBench: A Benchmark Suite for Fault-Tolerant Quantum Compilation and Architecture
   arXiv preprint arXiv:2601.03185
8. Shuwen Kan, Zefan Du, Chenxu Liu, **Meng Wang**, Yufei Ding, Ang Li, Ying Mao, and Samuel Stein
   SPARO: Surface-code Pauli-based Architectural Resource Optimization for Fault-tolerant Quantum Computing
   arXiv preprint arXiv:2504.21854
9. Fei Hua, Yuwei Jin, Ang Li, Chenxu Liu, **Meng Wang**, Yanhao Chen, Chi Zhang, Ari Hayes, Samuel Stein, Minghao Guo, Yipeng Huang, and Eddy Z. Zhang
   A Synergistic Compilation Workflow for Tackling Crosstalk in Quantum Machines
   arXiv preprint arXiv:2207.05751
10. Ang Li, Chenxu Liu, Samuel Stein, In-Saeng Suh, Muqing Zheng, **Meng Wang**, Yue Shi, Bo Fang, Martin Roetteler, and Travis Humble
    TANQ-Sim: Tensorcore Accelerated Noisy Quantum System Simulation via QIR on Perlmutter HPC
    arXiv preprint arXiv:2404.13184

WORKSHOP PUBLICATIONS

1. 🔗 **QCS '23** **Meng Wang**, Fei Hua, Chenxu Liu, Nicholas Bauman, Karol Kowalski, Daniel Claudino, Travis Humble, Prashant J. Nair, Ang Li
Enabling Scalable VQE Simulation on Leading HPC Systems
The International Conference for High Performance Computing, Networking, Storage, and Analysis Workshop: Quantum Computing Softwares, 2023

2. 🔗 **QCCC '23** **Meng Wang**, Bo Fang, Ang Li, and Prashant J. Nair
Efficient QAOA Optimization using Directed Restarts and Graph Lookup
The Second International Workshop on Quantum Classical Cooperative Computing, 2023

3. 🔗 **Q-CORE** Yue Shi, Chenxu Liu, Samuel Stein, **Meng Wang**, Muqing Zheng, and Ang Li
Design of an entanglement purification protocol selection module
Q-CORE: Quantum-Classical Orchestration for Resilient Engineering, 2025

WORK EXPERIENCE

Pacific Northwest National Laboratory *September 2024 - September 2025*
Ph.D. Research Intern. Mentor: Dr. Ang Li Richland, WA, USA

- Led research on architectural optimizations for Fault-Tolerant Quantum Computing (FTQC), resulting in an accepted paper at **ISCA'26** focused on transpiler-architecture co-design to curb Clifford costs, with follow-up work currently in preparation for **MICRO'26**.

Pacific Northwest National Laboratory *October 2022 - August 2023*
Ph.D. Research Intern. Mentor: Dr. Ang Li, Dr. Bo Fang Richland, WA, USA

- Participated in developing simulation frameworks leveraging leading HPC systems including Frontier, Summit, and Perlmutter to simulate large-scale quantum circuits on GPU clusters. The simulation framework is open-sourced at https://github.com/pnnl/NWQ-Sim.
- Led research improving the efficiency of classical optimization in Quantum Approximated Optimization Algorithm (QAOA), resulting in the publication of Red-QAOA (**ASPLOS '24**), which introduces a novel circuit reduction technique for efficient variational optimization.

TEACHING EXPERIENCE

Teaching Assistant at UBC Vancouver, BC, Canada
CPEN 312 - Digital Systems and Microcomputers *January 2024 - April 2024*
CPSC 322 - Introduction to Artificial Intelligence *July 2022 - August 2022*
APSC 160 - Introduction to Computation in Engineering Design *January 2022 - April 2022*
CPEN 411 - Computer Architecture *September 2021 - December 2021*
CPSC 340 - Machine Learning and Data Mining *January 2021 - April 2021*

AWARDS

Faculty of Applied Science Graduate Award *2021, 2022, 2023, 2024, UBC*
The annual award, subject to annual review and criteria fulfillment.

APSC Capstone Faculty Award *May 2020, UBC*

Dean's Honour List *Jun 2019, UBC*

SKILLS

Programming languages:
Verilog, C, C++, CUDA, HIP, MPI, Java, Python, ARMv7 Assembly, Racket.
Hardware:
System-on-Chip (SoC), FPGA.